

BEULAH AKINDELE

Principal Engineer, Payments Platforms | Lagos, Nigeria (Remote)

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PROFESSIONAL SUMMARY

Principal engineer with 5 years of experience, currently owning the Java and .NET services that move interbank transfers for 500+ financial institutions across multiple African countries, on a platform aggregating 10+ payment gateways behind three REST and SOAP contracts owned end-to-end for both internal and public consumers. Integrates straight into the switches those transfers depend on, across both ISO 20022 and ISO 8583, and designs for the moment one of them fails mid-transaction. Works the full depth of the money path, from the API contract and the message format through relational schema design and query tuning to what the service does on Kubernetes at 3am when an upstream switch starts timing out. Leads design reviews, roadmap and capacity planning, sets the engineering-quality standard, and mentors the team. Recent results: a 96% cut in reconfirmation queue depth, 38% fewer failed transactions week-on-week, and institution onboarding reduced by 99.75%, held at 99.9%+ uptime.

CORE COMPETENCIES

Payments & Financial Systems Interbank Transfers & Money Movement at Scale, ISO 20022 & ISO 8583 (transfers, mandates, reporting), Switch & Processor Integrations (10+ gateways), Core Banking Ledger Postings, Settlement & Reconciliation, Idempotent Transaction Processing, Wallet & Transfer Systems, PCI DSS Compliance

Architecture & Systems Event-Driven Microservices, Distributed Systems, RESTful API Design, Service Boundaries & Data Ownership, Idempotency, Circuit Breaker / Retry-with-Backoff, Multithreading & Asynchronous Processing, Caching Strategies

Languages Java, TypeScript, C#, JavaScript, Python, Go (hands-on, non-production), SQL, Shell Scripting

Databases PostgreSQL, MySQL, SQL Server, Oracle 19C: schema design, indexing, query optimization, transactions, stored procedures, zero-downtime migrations; MongoDB, DynamoDB, Redis, Elasticsearch; Hibernate, EF Core

Messaging & Streaming Kafka, RabbitMQ, Azure Service Bus (production: queues, scheduled-retry, dead-lettering), AWS SNS & SQS, BullMQ/Redis, Pub/Sub, Consumer Idempotency & Dead-Letter Reconciliation Patterns

Reliability & Observability High Availability, Resilience Engineering, Grafana Dashboards & Service-Health Metrics, Queue-Depth & Retry-Count Alerting, Production Debugging & Root-Cause Analysis, Zero-Downtime Deployment

Cloud & DevOps AWS (EC2, S3, RDS, Lambda, CloudFront, CloudWatch), Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions, Octopus Deploy, Argo CD), Git

Practices Technical Leadership, Design Reviews, Code Review & Engineering Standards, Mentoring, Agile/Scrum, Technical Documentation

EXPERIENCE

Principal Full Stack Software Engineer

March 2025 – Present

Qore

Java/Spring Boot, C#/.NET, Node.js/TypeScript, Azure Service Bus, Redis, PostgreSQL, SQL Server, Docker, Kubernetes, REST

- Design and ship backend microservices in **Java/Spring Boot and .NET** across a Channels platform serving **500+ financial institutions in 9 African countries**, owning API contracts, gateway middleware, and message-queue consumers that move interbank transfers end-to-end across **multiple payment gateways**. Replaced a 3-day cross-departmental onboarding process with guarded, API-driven validation forms, cutting new-institution onboarding to under 5 minutes (**99.75%** faster).
- Architected the platform's first **ISO 20022** integrations (NIBSS NPS, Interswitch IRIS) as **event-driven asynchronous services**: a synchronous acknowledgement returned inside the switch's SLA window while fulfilment dispatches over message queues, with **Redis-backed idempotency** keyed per message type so a retransmitted pacs.008 is rejected rather than double-posting a credit, and **exponential-backoff retry** across five attempts before **dead-lettering for manual reconciliation**, alerted on queue depth and retry-count distribution. Established the onboarding pattern every gateway has followed since, across three gateway upgrades and two new integrations at **zero downtime**.
- Re-architected transaction confirmation processing from shared to **gateway-filtered batching queues** after diagnosing a cross-gateway bottleneck behind a production incident, added **progressive retry throttling** (5, 10, then 15 minute intervals) and **Grafana gateway-health metrics** (confirmation volume, per-gateway success rate); drove a **96% reduction in reconfirmation queue depth**, **38% fewer failed transactions** week-on-week, and **99.9%+ uptime**.
- Eliminated silent billing failures by **decoupling billing from the KYC verification request path into a dedicated worker service**: a database-backed worker queue guarded by **retry-with-backoff** and a **circuit breaker**, letting the request path **degrade gracefully** when a third-party identity provider does, across a five-endpoint verification suite later extended with two new offerings.
- Debugged and resolved silent production failures across the stack: a schema constraint dropping audit-log writes for oversized third-party error payloads, fixed with a versioned stored procedure and backwards-compatible migration enabling **zero-downtime deploys**; a database performance degradation dropping inbound transactions, fixed with

scheduled archiving and query routing across live/archive tables; and a cyclical service-to-service call loop, resolved by relocating query handling into the owning gateway application.

- Led an architecture restructure sharpening service-responsibility boundaries: **module-based policy architecture** for gateway and fee rules, a **global exception handler**, and standardized thread-safe logging, explicitly targeting testability and maintainability. **Lead design reviews** and own engineering-quality standards, sprint planning, and release readiness, and **mentor engineers** toward a culture of ownership and quality.
- Led platform-wide security hardening across Docker-containerized services orchestrated on Kubernetes: **Zero Trust** inter-service auth with short-lived tokens, cryptographic transaction-integrity verification, encrypted credential storage, and PII log sanitization, isolating shared services onto dedicated containers to reduce blast radius while maintaining **PCI DSS compliance**.

Software Engineer & Calypso Technical Consultant

February 2024 – February 2025

Seaico Technologies Limited

Java, Spring Boot, Kafka, RabbitMQ, JavaFX, React.js, Oracle 19C, PostgreSQL, REST

- Led development of the Seaico Retail Platform on a **6-service Java/Spring Boot microservices architecture** with a real-time Calypso data-sharing pipeline, owning architecture decisions, REST API contracts, and code review across a **team of 4 developers**.
- Designed and built **TradeLink**, a YAML configuration-driven Java middleware synchronizing trade data across Calypso, BMatch, and IntegralFX at **sub-second propagation latency** over **Kafka and RabbitMQ** alongside SFTP, FTP, REST, and SMTP, with dynamic field-level transformation across XML, JSON, and CSV.
- Configured and tuned **Java 8 JVM runtime parameters** (heap sizing, GC settings) to each client's system constraints and workload profile across Calypso Fixed Income/FX deployments at Access Bank (Nigeria, Ghana, Cameroon), Zenith Bank (Nigeria & UK), and FCMB, owning DB provisioning and bespoke integration development on the banks' own infrastructure.
- Served as technical liaison in client meetings alongside product and project stakeholders, translating requirements from both technical and non-technical counterparts into validated, scoped project plans.

Machine Learning & Backend Engineer

February 2023 – January 2024

Buntu (FKA Get Step)

Python, Django, scikit-learn, NumPy, TensorFlow, TF-IDF, PostgreSQL, React.js

- Refactored Django services to delegate serialization upstream and **split recommendation and wallet logic into independent microservices: 450% performance improvement, 30% throughput increase, and 35% query-execution improvement** via stored-procedure rewrites and composite indexing chosen around real query patterns.
- Built wallet and transfer infrastructure integrating **Paystack and Providus** banking services, enabling auditable, secure financial actions across multiple channels.

Full Stack Software Engineer

October 2021 – January 2023

TBO Consulting Limited

React.js, Node.js, Express.js, MySQL

- Solely designed and delivered a Talent-as-a-Service platform (Figma through cloud deployment) with skill-based candidate matching and an end-to-end interview-scheduling workflow integrating Calendly and Zoom for automated booking, calendar sync, and meeting-link generation.

PERSONAL PROJECTS

TBO VisiGuard / VMS (visiguard.org)

React 19, TypeScript, Node.js, Express, Knex/MySQL, Socket.io, Docker, Nginx, AWS EC2/RDS/S3/CloudFront/SNS/SQS, GitHub Actions

Multi-tenant visitor-management SaaS architected end-to-end: **5-tier RBAC**, real-time Socket.io visitor events, scheduled overstay detection with SMS checkout links, and an **agnostic identity-provider sync layer** unifying Azure AD (MS Graph) and AWS IAM Identity Center behind one interface. Dockerized behind Nginx and shipped to AWS through a full **GitHub Actions CI/CD pipeline**.

Spartan / SPRTN.AI (sprtn.net)

NestJS, TypeScript, MongoDB, BullMQ + Redis, React 18, Next.js (App Router), Paystack

Live, revenue-generating multi-tenant SaaS giving each tenant an isolated backend for CMS, eCommerce, and real-estate listings, partitioned by companyId with per-tenant SMTP and payment-gateway configuration. **Modular service architecture, JWT access-refresh plus API-key** auth for public embed widgets, and **BullMQ/Redis-backed async jobs**.

EDUCATION

B.Eng. Electrical & Electronics Engineering, Covenant University